

Problem

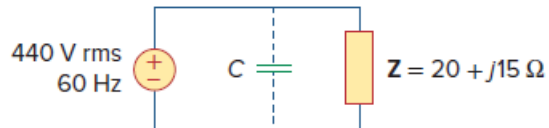
Refer to the circuit shown in Fig. 11.88.

(a) What is the power factor?

(b) What is the average power dissipated?

(c) What is the value of the capacitance that will give a unity power factor when connected to the load?

Figure 11.88:



Step-by-step solution

Step 1 of 4

Refer to Figure 11.88 in the text book.

Calculate the current passing through the impedance Z .

$$\begin{aligned} I &= \frac{440 \angle 0^\circ}{20 + j15} \\ &= \frac{440 \angle 0^\circ}{25 \angle 36.87^\circ} \\ &= 17.6 \angle -36.87^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I passing through the impedance Z is,

$$17.6 \angle -36.87^\circ \text{ A rms}.$$

Step 2 of 4

(a)

Calculate the value of the power factor.

$$\text{pf} = \cos(\theta_v - \theta_i)$$

Substitute 0° for θ_v and -36.87° for θ_i .

$$\begin{aligned} \text{pf} &= \cos(\theta_v - (-36.87^\circ)) \\ &= \cos 36.87^\circ \\ &= 0.8 \end{aligned}$$

Therefore, the value of the power factor is 0.8.

Step 3 of 4

(b)

Calculate the value of the complex power absorbed by the load.

$$S = VI^*$$

Substitute $17.6\angle -36.87^\circ$ A rms for I and $440\angle 0^\circ$ V rms for V .

$$\begin{aligned} S &= (440\angle 0^\circ)(17.6\angle -36.87^\circ)^* \\ &= (440\angle 0^\circ)(17.6\angle 36.87^\circ) \\ &= 7744\angle 36.87^\circ \\ &= 6195.19 + j4646.41 \text{ VA} \end{aligned}$$

Therefore, the value of the complex power is, $6195.19 + j4646.41$ VA.

Compare $6195.19 + j4646.41$ VA with $P + jQ$ where P and Q represents the average power and reactive power respectively.

Therefore, the value of the average power is 6195.19 W.

Step 4 of 4

(c)

For the circuit to operate at unity power factor, the value of the reactive power should be suppressed to zero by inserting a capacitor across the load.

Calculate the value of the capacitor that produces reactive power of 4646.41 VAR.

$$C = \frac{Q}{2\pi fV^2}$$

Substitute 4646.41 VAR for Q , 60 Hz for f and 440 V rms for V .

$$\begin{aligned} C &= \frac{4646.41}{2\pi(60)(440)^2} \\ &= \frac{4646.41}{2\pi(60)(193600)} \\ &= 63.66 \text{ }\mu\text{F} \end{aligned}$$

Therefore, the required value of the capacitor C to operate the circuit at unity power factor is,

$63.66 \text{ }\mu\text{F}$.